ICT GCE A Level 2020 - Paper 2

Duration: 2h30 | Cameroon GCE Board

Cameroon General Certificate of Education Board - General Certificate of Education Examination / Technical and Vocational Education Examination. Information and Communication Technologies, Paper 2 (0796), ADVANCED LEVEL, June 2020. Duration: two and a half hours.

Answer any SIX questions. All questions carry 17 marks each. For your guidance, the approximate mark for each part of a question is indicated in brackets.

You will be marked on your ability to use good English, to organize information clearly and to use specialist vocabulary where appropriate.

In calculations, you are advised to show all the steps in your working, giving your answer at each stage.

Exercise 1 (17 marks)

i. Explain the following computer terms giving an example for each.

(a) Automation. (2 marks)

(b) Control system. (2 marks)

(c) Robot. (2 marks)

ii. Discuss TWO factors that determine the reliability of an Information system. (2 marks)

iii. Identify TWO natural hazards that can affect a computer system and give a measure that can be used to secure the data in each case. (4 marks)

iv. (a) Distinguish between Simulation and Prototyping? (2 marks)

(b) List the steps used in prototyping. (2 marks)

(c) Give the role of end users' feedback in prototyping. (1 mark)

Exercise 2 (17 marks)

i. One method of data transmission is asynchronous.

(a) Describe asynchronous data transmission giving its disadvantages and advantages. (2 marks)

(b) Describe another method of data transmission: (1 mark)

ii. (a) List and explain THREE components of an Information System. (3 marks)

(b) Describe the following Information Systems, stating an appropriate use of each

- GIS
- DSS (4 marks)

iii. (a) What is a programming paradigm? (1 mark)

(b) Briefly define the following programming paradigms.

- Imperative (2 marks)
- Declarative (2 marks)

(c) Discuss briefly what is meant by high level programming language? (2 marks)

Exercise 3 (17 marks)

i. (a) Define System Development Life Cycle (SDLC). (1 mark)

(b) Name the SIX stages of SDLC and give TWO major tasks of the first three stages. (6 marks)

ii. The web form below is used to collect user input data:

```
<form action=*"user.php"* method=*"post"* >
<p>
Name:
<input type=*"text"* name=*"name"*/>
<p/>
<form/>
```

(a) Name the scripting language used to process the form above. (1 mark)

(b) What is the use of the attributes:

- action (1 mark)
- method (1 mark)

(c) Name another method value that can be used in place of *"post"*. (1 mark)

iii. (a) Discuss with examples THREE impacts of the use of ICT in health care in Cameroon. (3 marks)

(b) Give THREE reasons why the Health Information System is not very effective in Cameroon? (3 marks)

Exercise 4 (17 marks)

i. (a) What is system software? (1 mark)

(b) Describe the role of any TWO system software. (2 marks)

(c) Explain the *Round Robin* and the *First Come, First Serve* process management strategies used by the operating system in job scheduling. (3 marks)

ii. (a) Use a label diagram to describe machine instruction cycle of the computer. (2 marks)

(b) Explain the functions of THREE components of the CPU. (3 marks)

(c) Give the main function of the Random Access Memory (RAM) in the computer system. (1 mark)

iii. Differentiate between *Interrupt* and *Polling*. (2 marks)

iv. (a) What is a file extension? (1 mark)

(b) Copy and complete the table below in your answer booklet: (2 marks)

FILE TYPE	FILE EXTENSION
Bitmapped graphics	.JPEG
	.PDF
Video	

Exercise 5 (17 marks)

i.

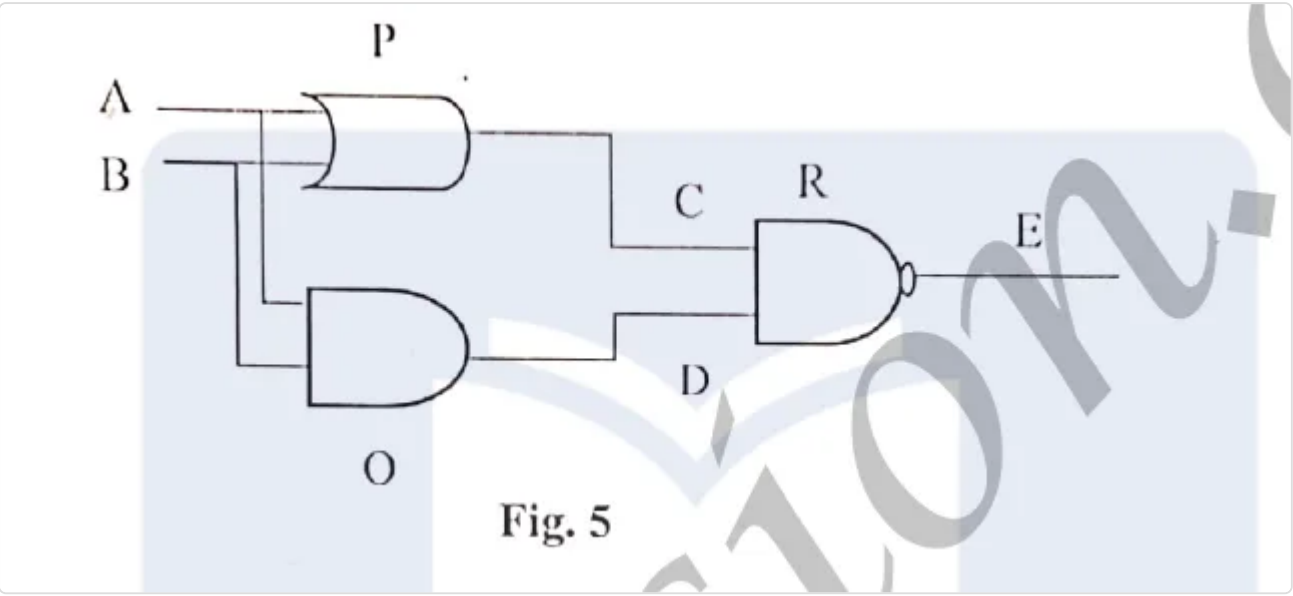


Fig. 5

(i) From the digital circuit above (Fig. 5), identify the logic gates labeled, P and R. (2 marks)

ii. Write out the Boolean expression of the outputs C, D and E in terms of inputs A and B. (3 marks)

iii. Construct a truth table for this digital circuit. (4 marks)

iv. (a) What is a project? (2 marks)

(b) Name and explain TWO methods used in controlling projects. (4 marks)

(c) Explain TWO responsibilities of a project manager. (2 marks)

Exercise 6 (17 marks)

i. CAL is a pedagogic approach gradually being used in Cameroon today.

(a) Define Computer Assisted Learning (CAL) (1 mark)

(b) Briefly discuss TWO course delivery methods used in CAL. (2 marks)

(c) Explain ONE advantage and ONE disadvantage of CAL in Cameroon. (2 marks)

ii. Define the following Database terms:

(a) Relationship. (1 mark)

(b) Attribute. (1 mark)

(c) Entity. (1 mark)

(d) Redundancy. (1 mark)

iii. (a) What is Normalization in relation to database modeling? (2 marks)

(b) When is a table in a database said to be in First Normal Form? (2 marks)

iv. In time-sharing multi-programming environment, many users share the same system simultaneously. This situation can result to various security problems.

(a) Identify TWO of such security problems. (2 marks)

(b) Differentiate between multitasking and multiprogramming. (2 marks)

Exercise 7 (17 marks)

i. Explain briefly the use of the following input technologies and cite a situation in which each can be useful:

(a) Bar code reader. (2 marks)

(b) Optical Character recognition. (2 marks)

(c) Automatic speech recognition. (2 marks)

ii. (a) Differentiate between data integrity and data verification. (2 marks)

(b) Name and describe TWO data verification techniques. (2 marks)

iii. (a) Explain TWO functions of a system analyst. (2 marks)

(b) Explain TWO ways a System Analyst can collect information about a system for analysis. (2 marks)

iv. (a) Convert the octal number 256 to binary (base 2). (1 mark)

(b) Convert the binary number 11011.11 to decimal (base 10). (2 marks)

Exercise 8 (17 marks)

i. Explain the following online Electronic Services below giving ONE advantage and ONE disadvantage for each:

(a) E-Government. (3 marks)

(b) E-Banking. (3 marks)

(c) E-Commerce. (3 marks)

ii. (a) What is a social network? Give one example. (2 marks)

(b) Outline ONE advantage and TWO disadvantages of social networking to students in Cameroon. (3 marks)

iii. (a) What is meant by the term IP address as it relates to the Internet? (1 mark)

(b) Briefly explain how protocols send messages across the Internet. (2 marks)

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